THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 13: **Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): CFCs are harmful to the environment. Reason (R): They release chlorine atoms in the stratosphere which destroy ozone. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A 100 W bulb is used for 5 hours a day for 30 days. If 1 kWh of electricity emits 0.8 kg of CO2, the emission due to the bulb is: 1

(a) 15 kg **(b)** 1.2 kg **(c)** 120 kg **(d)** 12 kg

3. Which is an example of a positive feedback in climate? 1

(a) melting ice lowers albedo and increases warming
(b) plants absorb CO2
(c) rain cools the air
(d) clouds reflect light

4. The Sun delivers about 1370 W per square metre to a surface facing it at the top of the atmosphere. Because the Earth is a sphere, the average over its whole surface is this value divided by: 1

(a) 1 **(b)** 4 **(c)** 2 **(d)** 3

5. Assertion (A): Nitrogen is abundant in the atmosphere but plants cannot absorb it directly. Reason (R): Plants need nitrogen in the form of compounds such as nitrates. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. The LED of 10 W replaces the 100 W bulb. The percentage energy saving is: 1

(a) 99 per cent **(b)** 90 per cent **(c)** 50 per cent **(d)** 10 per cent

7. Which of these is a negative feedback? 1

(a) burning forests
(b) more CO2 helps plants grow and absorb more CO2
(c) methane release
(d) melting ice

8. The average radiation per unit area (1370/4) is about: 1

(a) 342 W/m squared **(b)** 1370 W/m squared **(c)** 100 W/m squared **(d)** 685 W/m squared

9. Assertion (A): Rainwater is slightly acidic even without pollution. Reason (R): Atmospheric carbon dioxide dissolves in it. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. A hectare of forest holds 150 tonnes of carbon. The CO2 equivalent (multiply by 44/12) is about: 1

(a) 1500 tonnes **(b)** 400 tonnes **(c)** 150 tonnes **(d)** 550 tonnes

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. State two advantages and two limitations of afforestation to fight climate change. 2

12. How much CO2 is produced on burning 1 kg of carbon (C = 12, O = 16)? 2

13. Why is the sea-level rise caused by thermal expansion as well as melting? Explain. 2

14. How does the fall in the ozone level affect the phytoplankton and the food chain? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Calculate the mass of CO2 produced on burning 6 kg of carbon completely (C = 12, O = 16) and the mass of oxygen used. 3

16. Describe how weathering of rocks removes CO2 from the atmosphere over long time scales. 3

17. Explain the role of the phytoplankton in the oxygen and the carbon cycle. 3

OR

Explain how solar radiation heats the Earth's surface which in turn heats the atmosphere from below. 3

18. Compare the two gases CO2 and CH4 with regard to their sources and their heat-trapping power. 3

OR

Describe the impact of a decrease in the Arctic sea ice on weather, wildlife and shipping. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the Earth's energy budget with the greenhouse gases, the role of water vapour, CO2 and methane, and why the equilibrium temperature is higher than the temperature without the atmosphere. 5

OR

Prepare a plan for a carbon-neutral school: energy, transport, waste, water and green cover with measures, estimates and monitoring. 5

20. Analyse the carbon budget: emissions 10 Gt of carbon per year, 30 per cent absorbed by the oceans, 25 per cent by land plants, the rest stays. Find the quantities and the annual increase of the atmospheric carbon, and the concentration rise in ppm (1 ppm is about 2.1 Gt of carbon). 5

OR

Evaluate the statement 'Earth as a system: a change in one component affects all others' using the example of ocean warming, and trace the effects through the other three spheres. 5

END OF QUESTION PAPER — Total: 40 Marks